STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.	Case 19-G-0309
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service.	Case 19-G-0310

DIRECT TESTIMONY

OF

DANIELLE PANKO

Dated: August 30, 2019
Albany, New York

UTILITY INTERVENTION UNIT
DIVISION OF CONSUMER PROTECTION
NYS DEPARTMENT OF STATE
99 WASHINGTON AVENUE
SUITE 640
ALBANY, NY 12231-0001
www.dos.ny.gov

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I. INTRODUCTION AND OVERVIEW

Q. Would you please state your name and business address?

A. My name is Danielle M. Panko, and my business address is 99 Washington Avenue, Suite 640, Albany, NY 12231.

Q. By whom are you employed, in what capacity, and what are your professional backgrounds and qualifications?

A. I currently hold the position of a Utility Analyst with the Utility Intervention Unit (UIU) of the New York State Department of State's Division of Consumer Protection. I received a Bachelor of Science in Mathematics from the State University of New York at New Paltz in 2001 and a Master of Science in Electrical Engineering from the State University of New York at New Paltz in 2008.

From 2000 to 2001, I served as an intern with Central Hudson Gas and Electric Corporation (Central Hudson) located in Poughkeepsie, New York, in the Accounts Service Department and subsequently in the Electrical Engineering Department. From 2004 to 2007, I worked as a reliability engineer for Philips Semiconductors. From 2007 to 2012, I worked for Consolidated Edison Company of New York, Inc. (Con Edison) in the Rate Engineering Department as an Analyst, and later as a Senior Analyst, in the Gas Rate Design Section. I joined the UIU in 2012. My primary responsibilities include assisting with UIU's participation in Public Service Commission (PSC or Commission) proceedings, researching utility policy and regulatory related issues, and representing UIU during various utility-related

meetings and rate case negotiations. I have worked on a myriad of New York utility cases in addition to other rate and policy proceedings.

Q. Have you previously testified before the Commission?

A. Yes. I previously submitted testimony in Cases 13-E-0030 and 13-G-0031 (Con Edison), 14-E-0318 and 14-G-0319 (Central Hudson), 14-E-0493 and 14-G-0494 (Orange and Rockland Utilities, Inc. (Orange and Rockland)), 15-E-0283, 15-G-0284, 15-E-0285 and 15-G-0286 (New York State Electric and Gas Corporation (NYSEG) and Rochester Gas and Electric (RG&E)), 16-G-0058 and 16-G-0059 (The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) and KeySpan Gas East Corp. d/b/a National Grid (KEDLI)), 16-G-0060 and 16-G-0061 (Con Edison), 16-G-0257 (National Fuel Gas Company (National Fuel)), 17-E-0238 and 17-G-0239 (Niagara Mohawk Power Corporation d/b/a National Grid (National Grid)), 17-E-0459 and 17-G-0460 (Central Hudson), and 18-E-0067 and 18-G-0068 (Orange and Rockland).

Q. What is the scope of this testimony?

A. I will primarily comment on the Embedded Cost of Service (ECOS) studies, revenue allocation, and rate design proposed by KEDNY and KEDLI (together, the Companies), as set forth in the Direct Testimony of the Rate Design Panel (KEDNY) and the Direct Testimony of the Rate Design Panel (KEDLI). The absence of discussion of other topics in this testimony should not be construed as support for, or opposition to, the Companies' positions.

Q. How is your testimony organized?

A. This testimony has six sections. The first section is an introduction to the forthcoming testimony. Second, I summarize my recommendations. Third, I provide the background of this current proceeding, including a discussion of the Companies' previous rate cases. Fourth, I discuss issues with the Companies' gas cost of service studies. Fifth, I discuss revenue allocation issues. Finally, in the sixth section, I discuss a few gas rate design issues.

Q. Have you prepared any exhibits to be filed with your testimony?

A. Yes, Exhibit___(DP-1) and Exhibit___(DP-2) accompanies my testimony. Exhibit___(DP-1) contains my proposed revenue allocation process for both KEDNY and KEDLI. Exhibit___(DP-2) contains the Companies' responses to Information Requests (IRs) that I relied upon in preparing this testimony.

II. SUMMARY OF RECOMMENDATIONS

Q. Please briefly summarize your recommendations.

A. My recommendations are as follows:

A. COST OF SERVICE

First, I recommend the Companies utilize the results of both pro-forma gas ECOS studies filed in each of these cases as a guide to understanding the cost of providing service to each of the individual service classifications (SC). Second, in the next set of rate filings, I recommend the Companies again provide the results

of multiple gas ECOS studies for use as a guide to allocate costs. Specifically, the Companies should submit ECOS studies in which the distribution costs are classified 100 percent (%) as demand-related, as they did in these rate cases. This proposal is consistent with the ECOS models presented in some other New York utility rate cases and/or rate plans that have previously been approved by the Commission. Third, no less than two months prior to the intervenor direct testimony deadline, each of the Companies should invite all parties in its next rate case to attend a meeting (outside of its technical conference), at which the Companies would present an overview of the detailed data set forth in its filed gas cost of service studies, proposed revenue allocation process, and proposed rate design (similar to the process adopted in the Commission's Orders in Case 15-E-0050 *et al*.[1] and 18-E-0067 *et al*.[2]).

B. REVENUE ALLOCATION

I recommend the revenue allocation process be based on the output, or results, of the two types of ECOS studies filed in each of the utilities' rate cases. The Companies filed one pro-forma ECOS model that utilizes a minimum system methodology to classify distribution main costs as both customer-related and demand-related. The Companies' second pro-forma ECOS model is based on the

[1] *See* Cases 15-E-0050 *et al*., Order Adopting Terms of Joint Proposal to Extend Electric Rate Plan at Appendix A p. 20 (filed June 19, 2015).

[2] *See* Cases 18-E-0067 *et al*., Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans Attachment A November 9, 2018 Joint Proposal (issued March 14, 2019) (gas ECOS study discussed on pp. 43-44).

classification of distribution main costs as 100 % demand-related. While I do not necessarily agree with the minimum system method (which is the only type of ECOS study the Companies used for revenue allocation purposes in these cases), collectively the two ECOS models filed in these cases provide a range of possible outcomes that are consistent with past methodologies approved by the Commission in utility rate cases for revenue allocation purposes.

Furthermore, to avoid placing an excessive burden on any particular SC and to ensure that all SCs help to share some of the burden of the rate increase, I propose the Commission set boundaries around the amount of the base delivery revenue change applied to each SC during the revenue allocation process. For example, a potential boundary structure may include the following guidelines: no SC should receive a revenue change more or less than 25% to 50% of overall revenue increase and no SC receives a revenue decrease.

For illustrative purposes, Exhibit___(DP-1) shows the effects of applying boundaries of 0.75 and 1.25 (KEDNY) and 0.5 and 1.5 (KEDLI) times the system average percentage increase to the revenue allocation process in these rate cases. However, I recognize that somewhat different boundaries might be appropriate depending on the final revenue requirements in these cases. Thus, I recommend the Commission set specific boundaries around the amount of base delivery revenue change applied to individual SCs to reflect the magnitude of the delivery revenue increase that is ultimately adopted by the Commission in these proceedings.

For illustrative purposes, Exhibit___(DP-1) shows the results of utilizing the Companies' proposed tolerance band of plus or minus (+/-) 25% around the system average Rate of Return (ROR) to determine if SCs are under or overearning (or within the tolerance band) according to the results in the Companies' two filed ECOS models. I propose if the results of both ECOS models indicate the ROR for a SC is within the tolerance band (for example, within +/- 25% of the average ROR), then the SC should receive close to the system average delivery revenue increase over present rates. If both ECOS models confirm a SC is earning below the tolerance band, the class should receive an increase that is higher than average (but no more than the maximum increase established in the boundaries described above). If both ECOS models confirm a class is earning above the tolerance band, the class should receive an increase that is lower than average (but not less than the minimum increase). If the two ECOS models provide contradictory ROR results (*e.g.*, the one study shows the class earning below the tolerance band and the other study shows the class earning above the tolerance band), the rate class should receive a delivery revenue increase that is close to the system average increase. Additionally, while Exhibit___(DP-1) includes temperature controlled (TC) and interruptible (IT) customers in this revenue allocation process for illustrative purposes, I note that some utilities exclude TC/IT customers from ECOS studies[3] and instead adopt market-based

[3] *See e.g.*, Case 18-G-0068, Direct Testimony of Gas Rate Panel pp. 8-9 (filed January 26, 2018) (stating that the ECOS study analyzes Orange and Rockland's two firm classes and gas revenues from interruptible service are excluded from the ECOS study).

rates for those customers. An alternative approach to revenue allocation for those classes may be warranted depending on market conditions, or if those customers are not included in the ECOS studies.

C. RATE DESIGN

My testimony provides general recommendations for rate design that can be further explored, as appropriate, depending on the Companies' final revenue requirements. The starting point for my recommended rate design approach is my proposed revenue allocation process, which I used to assign costs to individual service classes (*e.g.*, setting a revenue target for each SC). However, I observe there are a variety of factors that may change the revenue target allocated to each SC, such as a change in the Companies' revenue requirements that requires more or less revenues to be collected from all SCs. The revenue target allocated to each SC together with the fixed monthly customer charge and portion of the revenues that flow through each of the volumetric rates (*e.g.*, increasing the ratio of tail-block to mid-block) can potentially have an impact on the optimal rate design for each SC. As such, my rate design recommendations are explanatory at this time, since my recommendation will be dependent upon the final revenue requirements, which will influence the magnitude of any potential changes to the customer charge and volumetric rates.

Residential Non-Heating – SC 1A

I recommend gradually moving away from the existing, steeply declining volumetric block rate design. The movement must be gradual to avoid rate shock to high usage customers. My recommendation for the amount of moderate change to be implemented will vary depending on several factors, such as if the Companies' residential SC 1A class revenues are reduced as per my revenue allocation proposals and/or if the Companies' overall revenue requirements are reduced. For the higher users (*e.g.*, customers who use greater than 500 therms per month as typical customers use 10-21 therms), the Companies should check to make sure these customers are in fact non-heating customers, and move them to the correct class, where applicable.

Residential Heating – SC 1B

The Companies' residential SC 1B heating rates should be a designed to more effectively encourage energy efficiency improvements. As such, the SC 1B heating rates should, at a minimum, be gradually moved away from a declining volumetric rate toward a flat rate. This recommendation contrasts with the Companies' proposal to continue a declining rate structure for SC 1B with a similar tail-block rate (Block 3) to mid-block rate (Block 2) ratio, because that approach makes no movement toward encouraging greater energy efficiency. The magnitude of my proposed change will be dependent upon the final revenue requirements.

Residential Bill Impacts

I recommend the residential bill impact results for KEDNY and KEDLI include additional information to account for customers at a wider range of usage levels. Residential bill impact results for both KEDNY and KEDLI (*e.g.*, Exhibit___(RDP-4)) should be evaluated for the entire range of usage levels of those classes (*e.g.*, usage up to 50,000 therms) and include the percentage of customers (including low income customers) within each usage level so parties can understand the range of bill impacts for those entire SCs. This is particularly helpful analysis for the parties to understand if the rate design deviates from the Companies' rate proposals.

Residential Targeted Help Towards Understanding How to Reduce Usage

I recommend the Companies' continue to provide an increased level of targeted assistance to residential customers with the highest monthly gas usage (including low income customers) to ensure they understand how they can reduce their usage.

III. GENERAL RATE CASE BACKGROUND

Q. Briefly discuss the Companies' previous rate cases, which were initiated in 2016.

A. In its Order Adopting Terms of Joint Proposal Establishing Gas Rate Plans, issued December 16, 2016, in Cases 16-G-0058 and 16-G-0059, the Commission approved rate plans that levelized the Companies' revenue increases over three years, starting in calendar year 2017. The Companies' rate plans included a

system average increase for most of the Companies' firm service classes (*id*. at p. 35). The rate increase for most firm KEDNY SCs was approximately 27% in Rate Year 1 and 16% in Rate Years 2 and 3. For KEDLI, the increase for most firm customers was approximately 18% in Rate Year 1 and 9% in Rate Years 2 and 3.[4]

Q. Prior to the December 16, 2016 Order in Cases 16-G-0058 and 16-G-0059, what was the Companies' recent history of increasing rates to customers?

A. It is my understanding the Companies' rates remained relatively stable or only slightly increased during the years leading up to the 2016 cases.[5] Additionally, KEDNY earned above its Commission approved ROR for several years. To address issues related to KEDNY exceeding its earnings sharing target for most of the years covered by its five-year rate plan (2008-2012), the Commission initiated Case 12-G-0544 (In the Matter of the Commission's Examination of the Brooklyn Union Gas Company d/b/a National Grid NY's Earning Computation Provisions and Other Continuing Elements of the Applicable Rate Plan). In the Commission's Order Adopting Terms of a Joint Proposal, issued and effective June 13, 2013, the Commission approved the Joint Proposal which extended the KEDNY's current rate plan with modifications including, but not limited to: reducing KEDNY's allowed return on equity (ROE) from 9.8% to 9.4%, increasing its common equity capital structure from 45% to 48%, and changes to the Company's

[4] *See* Cases 16-G-0058 and 16-G-0059, Order Adopting Terms of Joint Proposal Establishing Gas Rate Plans, Attachment 1, September 7, 2016 Joint Proposal at Appendix 3 (Schedules 2 and 3, KEDNY) and Appendix 4 (Schedules 2 and 3, KEDLI) (issued December 16, 2016).

[5] *See id*. at p. 2 ("Base rates have not been increased for KEDNY in over a decade and for KEDLI since 2008.").

deferral mechanisms, performance metrics, and revenue requirement. (*id*. p. 4). The Joint Proposal also stated that KEDNY's existing base delivery rates would not change for two additional years (except for previously authorized revenue neutral changes) (*id*.). In the Commission's Order Adopting Gas Rate Plans for Keyspan Energy Delivery New York and Keyspan Energy Delivery Long Island, issued and effective December 21, 2007 in Cases 06-G-1185 and 06-G-1186, the Commission increased base delivery revenues for KEDLI by $60 million in Rate Year 1 (2008) and left KEDNY's revenues unchanged, in part due to recognition of cost savings and efficiencies associated with the National Grid/KeySpan merger. (p. 64). This Order also established a multi-year rate plan that ensured nearly stable base delivery rates for all major categories of customers for at least five years.

Q. Can you also provide some background information concerning these current cases?

A. Yes. In the Companies' initial April 2019 base rate filing, KEDNY and KEDLI requested to increase their revenues by $236.8 million (19.3%) and $49.4 million (6%), respectively.[6] The Companies' revenue requests incorporate proposals related to: upgrading infrastructure and enhancing safety, improving customer service, and stimulating clean energy alternatives.[7] The Companies' state that

[6] Initial Rate Filing Cover Letter p. 1 note 1 (filed April 30, 2019) ("The revenue increases presented in these rate filings build off KEDNY and KEDLI's calendar year ("CY") 2019 revenues, including levelization revenues of approximately $108 million and $29 million, respectively."). *See also* Exhibit___(RDP-4), Schedule 2, page 2 for each Company.

[7] Initial Rate Filing Cover Letter pp. 2-3 (filed April 30, 2019).

efficiency initiatives and other efforts have reduced the revenue requirement by $50 million.[8] For purposes of developing SC revenue targets, KEDNY and KEDLI requested an overall delivery revenue increase of $237.2 million and $58.2 million which equates to (20.3%) and (7.8%) over pro-forma base delivery revenues at present rates, respectively.[9]

On July 3, 2019, the Companies submitted corrections and updates to their April 30, 2019 rate filings which resulted in a net decrease of $41 million to KEDNY's revenue requirement and $11 million to KEDLI's revenue requirement. These net decreases were based upon updates to the Companies' revenue forecast, capital investment plans and operations and maintenance expenses.[10] In June 2019, the Companies' filed supplemental testimony to discuss the revenue requirement impacts that customers may experience if the Transcontinental Gas Pipe Line LLC's Northeast Supply Enhancement (NESE) project is not available during the proposed rate plan period but the Companies did not propose to modify their April 30th filings.[11] The Companies' July 3, 2019 update filing appears to assume the NESE Project will go forward as planned.

If approved, the Companies' requested revenue increases would impact approximately 1.8 million natural gas customer accounts located in parts of New

[8] *Id*. at p. 2.

[9] *See* Exhibit___(RDP-4), Schedule 1 for each Company.

[10] Corrections and Updates Filing Letter p. 1 (filed July 3, 2019).

[11] Supplemental Testimony Filing Letter pp. 1-2 (filed June 11, 2019) (stating, "The net impacts to KEDNY and KEDLI's revenue requirement should the NESE project not go forward as planned are increases of $5.7 million and $15.2 million, respectively. The Companies are not proposing to modify their April 2019 filings at this time, as the expectation is that the NESE project will be approved.").

York City and Long Island.[12] The requested rate changes will potentially impact approximately 1,218,969 KEDNY residential accounts, of which 594,542 (48.8%) are SC 1A residential non-heating, and 624,427 (51.2%) are SC 1B residential heating. Similarly, these rate changes will impact approximately 543,240 KEDLI residential customers, of which 93,819 (17.3%) use gas for purposes other than heating (SC 1A), and 449,421 (82.7%) use gas for heating (SC 1B).[13] Most of the Companies' remaining accounts are small commercial customers. The Companies also serve a variety of other customers, including government accounts, large commercial and industrial customers, interruptible customers, and electric generators. Although relatively few in number, these other customers collectively receive a large fraction of the total gas volumes that are delivered over the KEDNY and KEDLI gas systems. For example, gas delivered to electric generators constitutes nearly 20% of KEDNY's overall system therm deliveries and nearly 43% of KEDLI's overall system therm deliveries yet these deliveries provide only 0.55% (KEDNY) and 3.17% (KEDLI) of the overall system revenues.[14]

Unlike the 2016 rate cases (16-G-0058 and 16-G-0059), where the Companies proposed uniform across-the-board percentage increases (31.5% for KEDNY and 26.7% for KEDLI) for the residential and small commercial SCs

[12] KEDNY serves approximately 1.2 million customers in the boroughs of Brooklyn, Queens, and Staten Island in New York City. KEDLI serves approximately 590,000 customers in Nassau and Suffolk Counties on Long Island and the Rockaways in New York City.

[13] Exhibit ___ (DP-2) (Company response to UIU IR No. 13 Attachment 1).

[14] See Exhibit___(DP-2) (Company response to UIU IR No. 23).

(KEDNY SC 1A and 1B, SC 2-1 and 2-2, and SC 3; KEDLI SC 1A and 1B, SC 2A and 2B, and SC 3), the Companies are now proposing varied revenue increases based on the Companies preferred ECOS methodologies. These ECOS results, and corresponding revenue allocation proposals, include the following service classes:

Table 1. Companies' Service Classes used during the revenue allocation process[15]

KEDNY Service Classes		KEDLI Service Classes	
SC 1A	Residential Non-Heat	SC 1A	Residential Non-Heat
SC 1B	Residential Heat	SC 1B	Residential Heat
SC DG	DG Residential Distributed Generation	SC DG	DG Residential Distributed Generation
SC 2-1	General Service Non-Heat	SC 2A	General Service Non-Heat
SC 2-2	General Service Heat	SC 2B	General Service Heat
SC 3	Multi Family	SC 3A	Multi Family Non-Heat
SC 4A	High Load Factor Service	SC 3B	Multi Family Heat
SC 4A-CNG	Compressed Natural Gas Service	SC 4	Interruptible Sales
SC 4B	Year-round AC Non-Residential	SC 9	Wholesale NGV
SC 5A	On System Large Volume Interruptible	SC 7	Interruptible Transportation
SC 6	Temperature Controlled	SC 12	Temperature Controlled Sales
SC 7	Seasonal Off-Peak	SC 13	Temperature Controlled Transportation
SC 21	Baseload DG	SC 15	High Load Factor
		SC 16	Year Round Space
		SC 17	Distributed Generation Sales

Q. Do you have concerns regarding the Companies' inclusion of any of these SCs in the revenue allocation process?

A. Yes. I have concerns with the Companies' proposal to include TC/IT customers in the ECOS studies and use the ECOS results to determine the revenues allocated

[15] Service Classes shown in blue were added to the Companies' ECOS studies and revenue allocation methodologies in the 2016 KEDNY/KEDLI rate cases.

to TC/IT customers. In the Companies' 2016 rate cases, some parties took different positions in testimony regarding the issue of whether to apply cost-based rates to these TC/IT classes or continue market-based rate design. However, when the Commission approved the Joint Proposal in those 2016 cases, it approved a transition to cost-based rates; TC/IT customers were included in the ECOS studies and assigned rates which resulted in bill impacts less than the otherwise applicable firm service.[16] If the Companies' approach to cost-based TC/IT customer rates is accepted again in these cases, and these TC/IT classes continue to be a part of ECOS studies, then they should be subject to the same boundaries and rules that are applied to other SCs.

Q. Please provide information regarding the potential residential customer impacts of the Companies' rate filing?

A. Based on the Companies' rate filings, the bills paid by typical residential heating[17] and non-heating[18] customers will increase as shown in the table below.

[16] *See* Cases 16-G-0058 *et al.*, Order Adopting Terms of Joint Proposal and Establishing Gas Rate Plans p. 48 (issued December 16, 2016) ("Sections IV.3.4 and V.3.4 adopt the rate design proposed by the Companies, rather than continuing the market-based rate design as recommended by UIU and Staff in testimony. Staff asserts, however, that the resulting rates would equate to approximately a 20% discount for TC customers and a 30% discount for IT customers off of the otherwise applicable firm service rates measured on a total bill basis, which is approximately equal to the discount levels from firm rates that Staff recommended for setting a cap on market based rates charged to non-firm customers in order to account for the additional cost these customers incur to maintain an alternative fuel source.") (citations omitted). *See also id*. at Attachment 1 p. 10 and p. 64.

[17] For the Companies' average residential heating assumptions (SC 1B), see the following exhibits: KEDNY Exhibit ___ (RDP-4) Schedule 4 p. 2 and KEDLI Exhibit ___ (RDP-4) Schedule 4 p. 3 (both exhibits assume usage of 83 therms a month for the average residential heating customer). Note that the Companies filed modified versions of these exhibits with their corrections and updates testimony. *See* KEDNY Exhibit___(RDP-4CU), Schedule 4 p. 2 and KEDLI Exhibit___(RDP-4CU), Schedule 4 p. 3.

[18] For the Companies' average residential non-heating assumptions (SC 1A) see the following exhibits: KEDNY Exhibit ___ (RDP-4) Schedule 4 p. 1 (KEDNY: 10 therms a month) and KEDLI Exhibit___(RDP-

Table 2. Companies' Proposed Residential Customer Bill Impacts

Typical Residential Customer	*Testimony Filed April 30th*		*Corrections and Updates Filed July 3rd*	
	Delivery	Total Bill	Delivery	Total Bill
KEDNY (Heating)	17.78%	11.99%	13.93%	9.61%
KEDLI (Heating)	6.29%	5.15%	14.77%	12.17%
KEDNY (Non-Heating)	25.53%	21.62%	20.79%	17.77%
KEDLI (Non-Heating)	12.97%	10.76%	8.64%	8.39%

As Table Two illustrates, the Companies revenue request has changed since its initial filing. Additionally, it is my understanding there may be revenue impacts if the NESE project does not move forward as planned that were not included in the Companies' July 3, 2019 updated revenue request.[19] I observe the final revenue requirements adopted in these proceedings may be different than the Companies' most recent July 3, 2019 proposal due to a variety of factors. Regardless of the revenue requirements ultimately adopted, the Companies' proposals for how to distribute their revenue requirements among customer classes (*e.g.*, revenue allocation and rate design proposals) are closely tied to decisions the Companies made when developing their ECOS studies. As such, I will discuss the Companies' ECOS study approach first and then turn to the proposed rate design and other issues.

4) Schedule 4 p. 1 (KEDLI: 21 therms a month). Note that the Companies filed modified versions of these exhibits with their corrections and updates testimony. See KEDNY Exhibit___(RDP-4CU), Schedule 4 p.1 and KEDLI Exhibit___(RDP-4CU), Schedule 4 p. 1.

[19] *See infra* p. 14 note 11 and accompanying text.

IV. COST OF SERVICE

A. *Background*

Q. Please provide a few brief introductory comments concerning ECOS studies.

A. An ECOS study is often used as a tool in utility rate proceedings to apportion rate base and operating expenses among the various customer classes. This data is used in conjunction with existing revenues to estimate the overall system ROR and the ROR earned from each service classification as defined in the study. Sometimes, the focus is on historical revenues; other times forecasted revenue increases for the Rate Year (pro-forma) are used to estimate the overall system ROR (similar to what the Companies proposed in these cases). Regardless of the specific approach that is used, the analyst must exercise discretion and judgment in apportioning costs. Some of these judgments have been the subject of much debate and controversy throughout the last 40 years or more.[20]

The end result of an ECOS study is to estimate the overall system ROR and individual service classification RORs. The ECOS study results are particularly significant because of their potential impact on revenue allocation – how the rate

[20] For example, the National Association of Regulatory Utility Commissioners Gas Distribution Rate Design Manual, published in June 1989 (NARUC Manual) discusses the general judgements that can be used in an ECOS model. *See e.g.*, National Association of Regulatory Utility Commissioners Gas Distribution Rate Design Manual (published June 1989) p. 20 (stating "[t]here is no scientifically correct method of making necessary allocations. A certain amount of judgement must be used in any cost of service study. Consequently, cost allocation studies should only be utilized as a general guide or as a starting point for rate design.") One particularly controversial judgement has been the decision regarding the assignment of customer costs, particularly whether costs associated with the distribution system should be included as customer costs; the NARUC Manual recognizes the 100% demand approach. (*id*. at p. 22-23) (noting "[t]he contra argument to the inclusion of certain distribution costs is that mains and services are installed to serve demands of the consumers and should be allocated to that function").

increases are assigned to the various customer classes. The ECOS results, or output, are sometimes also used to support aspects of the rate design – particularly the monthly customer charge – within individual SCs.

The remainder of this section of my testimony will primarily focus on the Companies' decision to use a minimum system approach in an ECOS study to classify some distribution main costs as customer-related, instead of classifying these costs as 100% demand-related.

Q. You mentioned that some of the judgments used within cost of service studies have been controversial. Can you elaborate on this point with some examples?

A. Yes. Some gas and electric utilities analyze a portion of their distribution costs using some variation of a minimum system or a statistically-based zero-intercept method – thereby classifying a portion of these distribution costs as customer related. Other utilities choose not to prepare this type of analysis – they do not classify or allocate any of the distribution costs based on the number of customers. Even utilities that use a minimum system or zero-intercept approach may not fully implement this controversial approach when developing their revenue allocation and rate design proposals.[21] In recent years, I have seen how controversial the judgments associated with the classification of costs as "customer related" can be in the context of various New York utility rate cases,[22] as well as the Value of

[21] For example, *see* Companies' proposed revenue allocation process in their 2016 rate cases. *See* Cases 16-G-0058 and 16-G-0059, Testimony and Exhibits of Rate Design Panel (Book 16 for both KEDLI and KEDNY) pp. 23-24 (filed January 29, 2016).

[22] *See e.g.*, Cases 16-E-0060 *et al*, Order Approving Electric and Gas Rate Plans pp. 44-45 (issued January 25, 2017) (noting "there can be reasonable disagreement over the preferred methods for

Distributed Energy Resources Working Group Regarding Rate Design (Matter 17-01277).

Q. Do utilities in New York consistently use the same approach to classify costs within the various FERC accounts that are set forth in ECOS studies?

A. No. New York utilities use a wide variety of different methods to classify and allocate costs. While not specifically gas related, the March 6, 2018 presentation by the Joint Utilities (including National Grid) entitled "VDER Rate Design Working Group Joint Utilities Presentations: ECOS Analysis," filed in Matter 17-01277 provides some useful insight into this diversity. While the Joint Utilities' March 6th presentation is useful, it does not provide much detail explaining how each utility determined its cost of service methodologies. In some cases, the ECOS methodology (including the allocators) may have been adopted in response to intervenor testimony, or to comply with a negotiated memorandum of understanding,[23] stipulation agreement,[24] or past joint proposal. The presentation

undertaking, and the assumptions underlying an ECOS study."). See also, Cases 18-E-0067 *et al*, Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans p. 77 (issued March 14, 2019) (noting that two parties took issue with the utility's ECOS study methodology).

[23] *See e.g.,* Cases 16-E-0060 *et al*., Exhibit 9 of UIU Electric Rate Panel on the Joint Proposal (October 13, 2016). This discovery response provided information regarding a collaborative process "to resolve issues associated with the methodologies to be utilized by Con Edison in developing…" ECOS studies. In 2006, eight parties signed an MOU to support, or not oppose, the Company's use of two inputs in future ECOS methodologies.

[24] *See e.g*., Case 10-E-0050 (Niagara Mohawk) Exhibit 393. Several parties negotiated an agreement called the Rate Design, Customer and Market Issues Stipulation (2010 RDCM Stipulation or Stipulation). A 2017 discovery response provided a copy of the 2010 stipulation. *See* Case 17-E-0238 *et al*., Exhibit____(URP-1) (filed August 25, 2017) pp. 1-9. (The Stipulation was "intended to amicably resolve certain matters" to reduce the issues in conflict at the litigated evidentiary hearing and the terms were submitted as an "integrated whole" (p. 7), meaning the parties adopted the Stipulation in total and could not selectively adopt some sections. The Stipulation covered a variety of controversial issues,

does confirm a lack of uniformity in utility ECOS methodology and results. It appears ECOS methodologies (particularly as they relate to the concept or quantification of "customer-related" costs) have raised concerns among many of the non-utility parties in the VDER Rate Design Working Group in Matter 17-01277. To my knowledge, a similar comparison for gas utilities has not been developed in any recent generic proceedings. However, I provide my own review of some gas utility ECOS methodologies below.

Q. Why does it matter what judgments and datasets are used in a utility cost of service study?

A. These judgments and datasets can impact the final RORs for individual service classifications. Even though these RORs are the result of controversial judgments, they largely determine whether classes are deemed to be under-earning or over-earning during the revenue allocation process. In turn, this determines how large a rate increase will be assigned to each of the service classifications.

Q. Should the Companies' preferred ECOS results be used as the sole basis to allocate the revenue requirement for each utility across service classifications?

A. No. Based on my experience reviewing ECOS models and revenue allocation proposals in various New York utility rate cases over the past few years, I believe it is appropriate to start incorporating several ECOS models as a guide to allocate costs. This provides more insight into the impact of key judgments, it will help the

such as: (1) cost of service, (2) revenue allocation, (3) authorized deferrals, (4) additional flex rates, and (5) hourly pricing).

Commission make better, more informed decisions based on the results, and it will ameliorate, to some degree, the controversy that surrounds judgments concerning the appropriate definition and interpretation of so-called "customer-related" costs.

Q. Have the Companies previously agreed to file the results of using more than one ECOS model?

A. Yes. The Joint Proposal submitted in the KEDLI and KEDNY rate cases 16-G-0058 and 16-G-0059 on September 7, 2016, included a commitment to file certain ECOS studies in these proceedings. Page 131 of that Joint Proposal states:

> **In their next base rate cases, the Companies will each submit an embedded cost of service study in which the costs of mains will be classified 100 percent as demand related.** This requirement does not require the Companies to support using the results of this study for any purpose. Nor will it prohibit or prejudice the Companies in any way from presenting other embedded cost of service studies or making other revenue/costs allocation or rate design proposal in any future base rate case. (emphasis added)

The commitment regarding KEDNY and KEDLI's future gas ECOS studies was approved by the Commission in the "Order Adopting Terms of Joint Proposal and Establishing Gas Rate Plans" issued and effective December 16, 2016. The Companies filed the 100% demand studies in these cases[25] but they chose not to use the results.

[25] For more information regarding the Companies' filed ECOS models applying a 100% demand input for the cost of distribution mains, see the following exhibits and supporting workpapers. *See* KEDNY and KEDLI Exhibit___(RDP-12).

Q. Is this agreement unique?

A. No. Similar agreements can be found in the joint proposals for several utilities including (1) Orange and Rockland and (2) NYSEG and RG&E. The Orange and Rockland Joint Proposal requires the utility to file a few different electric and gas ECOS studies in the Company's next rate cases, including an ECOS study that classifies certain distribution accounts as 100% demand-related.[26] This provision was approved by the Commission in the "Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans" issued and effective March 14, 2019. The Joint Proposal filed in the NYSEG and RG&E electric rate cases 15-E-0285 *et al*. on February 19, 2016, also requires the Companies to file a few different electric ECOS studies in their next rate cases.[27] The Joint Proposal was approved by the Commission in the "Order Approving Electric and Gas Rate Plans in Accord with Joint Proposal" issued and effective June 15, 2016.[28] NYSEG and RG&E filed electric ECOS results consistent with this provision in case 19-E-0378 *et al*. on May 20, 2019.

An example of where a New York utility filed multiple types of ECOS studies in its rate cases can be found in Central Hudson's 2017 electric and gas rate cases 17-E-0459 and 17-G-0460. Central Hudson presented three different ECOS studies: 1) historical, 2) historical delivery only, and 3) pro-forma (forecasted) for

[26] *See e.g*., Case 18-E-0067 *et al*., Order Adopting Joint Proposal, Attachment A November 9, 2018 Joint Proposal (issued March 14, 2019) (gas ECOS study discussed on pp. 43-44).

[27] *See* 15-E-0283 *et al*., Joint Proposal at Appendix W (filed February 19, 2016).

[28] *See* 15-E-0283 *et al*., Order Approving Electric and Gas Rate Plans in Accord With Joint Proposal (issued June 15, 2016).

both gas and electric services.[29] For several service classifications, the study results, or the individual class ROR used to determine revenue allocation, differed significantly depending on which type of study was used (*i.e*., historical or pro-forma). For example, in one ECOS study a service classification was over-earning while in another study that same service classification was under-earning.[30] This disparity of results revealed cost causation was not easily and explicitly identified and assigned to individual service classifications. In recognition of the varying ROR results for most service classes among the ECOS models, Central Hudson recommended that most service classifications receive the same overall system average percentage rate increase.[31]

Q. Have any New York utilities made it difficult to learn the impact of classifying certain distribution costs as 100% demand-related rather than classifying some of these costs as customer-related?

A. Yes. An example can be found in a utility's previous rate cases (18-E-0067 and 18-G-0068). The "Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans" filed on October 16, 2015 approved the Joint Proposal, which modified the gas ECOS study in that case to classify distribution

[29] *See* Cases 17-E-0459 and 17-G-0459, Direct Testimony of the Central Hudson Cost of Service Panel pp. 20-22 (filed on July 28, 2017) (discussing results of its filed electric and gas ECOS studies).

[30] *See* Cases 17-E-0459 and 17-G-0460, Forecasting and Rates Panel Exhibit___(FRP-11) Schedule A: Electric Interclass Revenue Allocation (filed on July 28, 2017) (illustrating the Company's revenue allocation proposal).

[31] *See* Cases 17-E-0459 and 17-G-0460, Direct Testimony of the Forecasting and Rates Panel p. 43 (filed July 28, 2017).

gas mains as 100% demand-related and 0% customer-related.[32] Yet, in a subsequent rate case discovery process, the utility refused to provide gas ECOS results based on this classification approach.[33] To ensure that all parties have access to a variety of ECOS models, it is important a 100% demand-related ECOS study is included in a utility's filed rate case. Regardless of the preferences of the utility, it is useful to have the results of more than one approach to cost allocations, and it is reasonable to require the utility to provide these results for consideration by intervenors and the Commission, even if the utility disagrees with the methodology.

Q. Are there other cases in New York where the Department of Public Service (DPS) Staff or the utility proposed treating distribution gas mains as 100% demand-related?

A. Yes. For example, in cases 19-G-0379 (NYSEG) and 19-G-0381 (RG&E), these New York utilities proposed to classify distribution gas mains in FERC account 376 as 100% demand-related.[34] In cases 15-G-0284 (NYSEG) and 15-G-0286

[32] *See* Cases 14-E-0493 and 14-G-0494, Order Adopting Terms of Joint Proposal and Establishing Electric Rate Plan Attachment A Joint Proposal p. 45 (issued October 16, 2015).

[33] *See* Cases 18-E-0067 and 18-G-0068, Exhibit___(DP-1) at p. 1 (filed May 25, 2018). The Company response to UIU-2 IR No. 32 stated, "In Case 14-G-0494, the Company's filed ECOS study employed a minimum system methodology to classify distribution gas mains as having both a demand component and a customer component. The Company later agreed to a 100% demand related classification of distribution gas mains as part of the Joint Proposal, which was adopted by the Commission in its Order Adopting Terms of Joint Proposal, issued October 16, 2015, in Case 14-G-0494. The Company, however, does not agree with this methodology and is under no obligation to perform an ECOS study employing this methodology. Accordingly, the Company employed a minimum system methodology in classifying distribution gas mains in the ECOS study it filed in this proceeding. The Company objects to performing an ECOS study classifying distribution gas mains as 100% demand related."

[34] *See* Cases 19-G-0379 and 19-G-0381, Direct Testimony of David A. Heintz p. 27 (filed May 20, 2019).

(RG&E), both of these New York utilities proposed to classify distribution gas mains in FERC account 376 as 100% demand-related.[35] In those 2015 gas cases, DPS Staff did not appear to object to NYSEG and RG&E's ECOS studies which classified distribution gas mains as 100% demand-related.[36] Similarly, NYSEG and RG&E classified 100% of distribution gas mains as demand-related in several earlier gas proceedings, including cases 09-G-0716, 09-G-0718, and 01-G-1668.[37] In KEDNY and KEDLI rate cases 06-G-1185 and 06-G-1186, DPS Staff advocated to allocate distribution gas mains costs as 100% demand-related.[38]

Q. Are there other New York proceedings where similar cost of service issues are being discussed?

A. Yes, cost allocation issues are being discussed by the VDER Rate Design Working Group in Matter 17-01277. Multiple parties have asked questions regarding the electric cost of service models used by the various New York electric utilities, with a focus on how these models impact mass market customers. In the first half of 2018, some parties submitted IRs on this topic, including UIU. However, the responses UIU received provided very limited answers to its questions.[39] At this time, it is not clear how discussions in the working group will impact future cost of

[35] *See* Cases 15-G-0284 and 15-G-0286, Direct Testimony of David A. Heintz p. 27 (filed May 20, 2015).

[36] *See* Cases 15-G-0284 and 15-G-0286, Staff Gas Sales and Revenue Panel p. 48 (filed September 2015).

[37] *See* Cases 09-G-0716 and 09-G-0718, NYSEG and RG&E Direct Testimony of the Embedded Cost of Service Panel p. 11 and pp. 35-36 (filed September 17, 2009).

[38] *See* Cases 06-G-1185 and 06-G-1186, Direct Testimony of Aric J. Rider (filed January 2007) (p. 15).

[39] *See e.g.*, Matter 17-01277, Niagara Mohawk, Responses to UIU-1 to UIU-8 (filed March 29, 2018).

service models, revenue allocation, and rate design. For now, the most appropriate forum to discuss cost of service-related issues continues to be utility rate cases.

Q. Has the minimum system approach been universally accepted by either utilities or regulators?

A. No. This approach is not universally used, and where it has been used it has often been very controversial. Even when it has been used, it has not necessarily been fully relied upon (see for example, the KEDNY and KEDLI 2016 rate cases previously discussed in this testimony, in which the final revenue allocation was not entirely driven by the ECOS results).

Q. Has the Companies' consultant, Mr. Gorman, been consistent in always proposing to classify distribution mains as partly customer-related?

A. No. I am aware of some Pennsylvania gas utility proceedings in which Mr. Gorman allocated none of the distribution main costs as customer-related; instead these costs were classified as demand-related and the costs were allocated using the average and excess method.[40]

Q. Do you have comments regarding the Companies' choice to use pro-forma ECOS studies, instead of historical ECOS studies, or both?

[40] *See* Case 17-E-0238 *et al*., UIU Rate Panel Exhibit 1 pp. 32-33 (IR No. UIU-6 KOH-109) (filed August 25, 2017).

A. Yes. The Companies prepared historic ECOS studies in response to DPS IR No. 504 (which were not filed as part of the Companies' initial testimony). In Exhibit___(DP-2) (Companies response to UIU IR No. 20 part C), the Companies noted there were no major differences among the historical and pro-forma studies. Additionally, the Companies noted the revenue allocation would not change for KEDLI SCs and would only slightly change for KEDNY SC 3 (moving closer to the average ROR). I have not incorporated the historical ECOS models into my proposed revenue allocation process in these cases since the results are largely redundant; while there are some small differences, there are no extreme differences, so I see no reason to introduce the added complexity of working with both historical and pro-forma ROR results.

B. ***Recommendations***

Q. What do you recommend with respect to using ECOS results for revenue allocation purposes in these cases?

A. I recommend using both the results of the pro-forma gas ECOS studies which classify distribution main costs as 100% demand-related and the results of the ECOS studies proposed by the Companies, which use the minimum system approach. While I do not necessarily agree with the ECOS model that classifies distribution main costs as partly customer-related, I recognize the Companies appear to prefer this approach. The Commission has sometimes accepted ECOS results using a minimum system approach, just as it has sometimes accepted results in which distribution mains were classified as 100% demand-related. In the

context of this case, I think it is a reasonable compromise to use both models. This provides an overall range of possible outcomes consistent with the range of methodologies approved by the Commission or otherwise utilized in other rate cases for revenue allocation purposes.

Q. What do you recommend with respect to ECOS models for revenue allocation purposes in future cases?

A. First, I recommend the Companies provide the results of multiple gas ECOS studies for use as a guide to allocate costs to individual service classifications, including (1) pro-forma ECOS studies with distribution accounts classified as 100 percent demand-related, (2) pro-forma ECOS studies with distribution account classified as partly customer-related as well as demand-related, and (3) historical ECOS studies corresponding to (1) and (2) if the results are far different from the pro-forma results. This is consistent with the approach that has been accepted by some other New York utilities in their rate cases, and/or language that was included in Joint Proposals that have recently been approved by the Commission.

Furthermore, to help parties understand the results of the various ECOS models and the potential impact on rates, the Companies should invite all parties in the next rate cases to attend a meeting where the parties and their experts can review the detailed data set forth in its cost of service studies, the proposed revenue allocation process, and the proposed rate design. This meeting should be scheduled at least two months prior to the intervenor testimony deadline. The presentation should mirror the requirements approved by the Commission in the

recent Orange and Rockland and prior Con Edison Rate Plans[41] – in addition to other details that might be deemed necessary in this context.

Q. Do you have any concerns with the use of ECOS studies to allocate costs to TC and IT customers?

A. Yes. I have concerns regarding the Companies' proposal to use ECOS results to determine the revenues allocated to TC and IT customers. I note that some utilities do not use an ECOS study to design their interruptible rates.[42] Further, excluding these customers from the ECOS study gives the Companies more freedom to design IT/TC rates. However, if the Companies' proposed approach to cost-based IT/TC customer rates is accepted and these classes continue to be a part of ECOS studies, then TC/IT classes should be subject to the same boundary and rules applied to other SCs.

[41] *See e.g.,* Case 18-E-0067 *et al*., Joint Proposal (Orange and Rockland) p. 44 (filed November 9, 2018), ("Following its next gas rate filing, the Company [Orange and Rockland] will conduct, for interested parties, a walk-through of the ECOS study and rate design underlying the proposed gas base delivery rates. Additionally, the Company will provide and review at the walk through, an explanation of the differences in the ECOS studies filed pursuant to this Proposal, a more detailed explanation of the purpose of each file and cross-references of the underlying data sources, a table of acronyms used, a table of contents, and an index of files."). This provision has been approved by the Commission in the "Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plans" issued and effective March 14, 2019. *See also* Case 15-E-0050 *et al*., (Con Edison) Order Adopting Terms of Joint Proposal to Extend Electric Rate Plan at Appendix A p. 20 (filed June 19, 2015) (noting that "The 2013 Joint Proposal also provided that for the Company's [Con Edison] next electric rate filing, the Company will . . . (ii) conduct, for interested parties, a post-filing walk-through of the ECOS study and rate design underlying the proposed electric base delivery rates; and (iii) provide a more detailed explanation of supporting ECOS and rate design work paper documentation, which will include a process flow chart(including a basic explanation of the purpose of each file and cross-references of the underlying data sources), a table of acronyms used, a table of contents and index of files . . . all are applicable to the Company's next electric rate filing.").

[42] *See supra* p. 7, note 3. *See also*, Case 16-G-0061, Direct Testimony of Gas Rates Panel pp. 7-8 (filed January 29, 2016).

IV. REVENUE ALLOCATION

A. *Background*

Q. Can you briefly explain the revenue allocation process in a typical utility rate case?

A. Yes. The results of one or more ECOS studies guide decisions regarding how proposed rate changes should be allocated across various service classifications. The specific process can vary, but ECOS results are generally examined to see whether certain classes have been identified as having a high ROR (paying more than the costs they have been allocated – sometimes referred to as "overearning"). If a high ROR is identified, an effort is typically made to assign a somewhat smaller share of the rate increase or revenue requirement to that service classification, thereby moving in the direction of greater uniformity in the service classification RORs. Similarly, if certain classes are identified as having a low ROR (paying less than the costs that have been allocated to them, which is sometimes described as "underearning"), a somewhat greater than average share of the rate increase or revenue burden may be assigned to that class during the revenue allocation step.

In carrying out the revenue allocation process, tolerance bands are often used around the system average ROR to decide whether the service classification ROR deviates far enough from the average ROR to merit special treatment. The width of the tolerance band (*e.g.*, +-10%, 20%, 30%, etc.) can vary widely, but the effect is generally the same – to ameliorate disputes concerning the ECOS results,

since service classifications that fall within the tolerance band generally receive the same percentage change.[43]

An analyst must exercise judgment when determining the size of the tolerance band, and this is sometimes a topic of dispute. The tolerance band can be important because it helps determine which service classifications will receive revenue changes that are less than, or more than, the other service classifications. Finally, some restrictions are typically imposed on how much any service classification will deviate from the average (*e.g.*, no class is given a decrease, no class is given more than 150% of the average increase).

The typical process just described is not the only method that has been used in New York. Over the years in various proceedings, the Commission has sometimes approved other revenue allocation methods. For example, in the last KEDNY and KEDLI gas rate cases (16-G-0058 and 16-G-0059) the Commission approved system average increases for most of the Companies' service classifications without regard for the ECOS results. In those cases, the Commission stated in the Order Adopting Terms of the Joint Proposal and Establishing Gas Rate Plans on December 16, 2016:

> Pursuant to JP Sections IV.3.2 and V.3.2, the revenue increases in each of the rate years would be allocated to all firm service classes [service classifications] on an equal percentage basis to current delivery revenues,

[43] I note that note that Con Edison and Orange and Rockland's revenue allocation process is unique from other New York utilities since these utilities have in past testimony proposed to apply 1/3 of the adjusted surplus and deficient costs "in a revenue neutral manner prior to applying the revenue increases." *See e.g.*, Case 19-G-0061, Direct Testimony of Gas Rates Panel pp. 33-34 (filed January 29, 2019). *See also*, Case 18-G-0068, Gas Rates Panel pp. 23-24 (January 26, 2018).

> with limited exceptions…. Revenue increases for Rate Years 2 and 3 will be similarly allocated. (p. 35).
>
> And:
>
> We agree that given the sizeable revenue increases for KEDNY and KEDLI, the equal percentage allocation of the revenue increases to all firm service classifications would be appropriate to minimize rate shock to the extent practicable. (p. 38).

Q. Please explain how the Companies propose to distribute the 20.3% increase for KEDNY over pro-forma base delivery revenues, and the smaller revenue increase for KEDLI (7.8% increase over pro-forma base delivery revenues) among the various customer service classifications?

A. The Companies started with the results of its chosen pro-forma gas ECOS Study (which classifies distribution mains as partly customer-related using a minimum system methodology[44]), using the following steps: (1) applying a +-25% tolerance band around the system average ROR (service classifications above or below the +-25% tolerance band were considered surplus or deficient, respectively); (2) assigning revenue changes based on the tolerance band results (generally, classes within the tolerance band receive an average increase, classes over the band receive smaller increases or a decrease, and classes under the tolerance band receive larger increases).[45] The tables below, from the Companies technical

[44] The resulting classification of distribution costs from the minimum system is 47.08%/52.92% customer/demand-related (KEDNY) and 46.17%/53.83% (KEDLI) customer/demand-related.

[45] *See* Cases 19-G-0309 and 19-G-0310, KEDNY and KEDLI Direct Testimony of the Rate Design Panel pp. 26-27 (filed April 30, 2019).

conference presentation on May 30, 2019, display the Companies' initial proposed revenue allocation for each utility.

Table 3. KEDNY's Proposed ECOS and Revenue Allocation Results

Results of ECOSS and Revenue Allocation
KEDNY – Exhibit RDP-4, Schedule 1

KEDNY	RR at Current Rates	Current Base Delivery $M	Base Delivery Increase $M	Base Delivery % Increase
System Average	3.45%	$1,169	$237	20.3%
SC1A (Res NonHeat)	(1.5%)	$164	$46	28.2%
SC1B (Res Heat)	3.6%	$643	$128	19.9%
SC1DG (Res DG)	(7.2%)	<$0.1	<$0.1	28.2%
SC2-1 (Comm NonHeat)	20.6%	$61	$3	5.0%
SC2-2 (Comm Heat)	4.1%	$101	$20	19.9%
SC3 (Multi Fam)	1.3%	$134	$38	28.2%
SC4A (HLF)	11.1%	$13	$0.7	5.0%
SC4A (CNG-NGV)	16.6%	$1.5	<$0.1	5.0%
SC4B (Year-round AC)	19.3%	$1.2	<$0.1	5.0%
SC6 (TC-IT)	31.7%	$49	$1.0	2.0%
SC7 (Seasonal)	(10.6%)	<$0.1	<$0.1	28.2%
SC 21 (DG)	(15.8%)	$0.7	$0.2	28.2%
Note- Information reflects roll-in of Tax Credit and ETIP Surcharge				
WITHIN Band		*OVER Band*		*UNDER Band*

Table 4. KEDLI's Proposed ECOS and Revenue Allocation Results

Results of ECOSS and Revenue Allocation
KEDLI – Exhibit RDP-4, Schedule 1

KEDLI	RR at Current Rates	Current Base Delivery $M	Base Delivery Increase $M	Base Delivery % Increase
System Average	5.72%	$746	$58	7.8%
SC1A (Res NonHeat)	(0.5%)	$39	$6	15.6%
SC1B (Res Heat)	4.8%	$462	$37	8.0%
SC1DG (Res DG)	(4.0%)	<$0.1	<$0.1	15.6%
SC2A (Comm NonHeat)	23.5%	$52	$3	4.9%
SC2B (Comm Heat)	6.7%	$153	$12	7.6%
SC3A (Multi Fam Non)	13.1%	$1	<$0.1	4.9%
SC3B (Multi Fam Heat)	6.6%	$21	$1.6	7.6%
SC9 (NGV)	62.3%	$4	($1.4)	(36.0%)
SC4-7-12-13 (TC-IT)	40.7%	$3	$0.1	4.9%
SC15 (HLF)	25.2%	$4	$0.2	4.9%
SC 16 (Year Round)	6.6%	$8	$0.6	7.6%
Note- Information reflects roll-in of Tax Credit and ETIP Surcharge				
WITHIN Band		*OVER Band*		*UNDER Band*

B. ***Recommendations***

Q. What is your response to the Companies' revenue allocation proposals?

A. The Companies' proposed revenue allocation gives too much weight to the Companies' ECOS results based on the minimum system approach (exclusively focusing on these RORs to determine if individual classes are underearning/overearning). Also, some of the proposed changes in delivery rates are too extreme. For instance, KEDLI is proposing a revenue decrease of 36% for SC9 NGV and an increase of 15.6% (two times the system average of 7.8% equating to 15.6%) for SC 1A residential non-heat and SC 1DG residential Distributed Generation, as shown in the base delivery increase in the table above.

I recommend the revenue allocation process consider the output, or results, of both of the Companies' filed pro-forma ECOS studies (including the study that allocates distribution main as 100% demand-related).[46] While I do not necessarily agree with the minimum system method, I don't object to giving it some consideration in these proceedings, since the combination of both ECOS models provides a range of possible outcomes consistent with approaches that have been approved by the Commission or utilized by other New York utilities for revenue allocation purposes in prior rate cases. Furthermore, I propose that boundaries be placed around the amount of base delivery rate change applied to each rate class in order to avoid placing an excessive burden on any class, and to ensure that all

[46] To see which classes are under/over earning according to the two filed ECOS models in each case, I utilized the Companies' Exhibit___(RDP-4) and Exhibit___(RDP-12) (ECOS 100% demand study) workpapers.

the service classes help share some of the burden of the rate increase. For example, a potential boundary structure might include the following guidelines: no SC should receive a revenue change more or less than 25% to 50% of the overall increase and no SC should receive a revenue decrease.

For purposes of providing a complete illustrative example of a revenue allocation proposal for both KEDNY and KEDLI, I incorporated the Companies' proposed tolerance band of +/- 25% around the system average ROR. In other words, every SC with a calculated ROR falling within the tolerance band would be treated the same as every other SC that also falls within the tolerance band. Each of these rate classes should receive the same percentage increase – a percentage factor that is fairly close to the overall system average delivery revenue increase relative to present rates. If the two ECOS studies yield contradictory conclusions (*e.g.*, one study shows a class as earning below the tolerance band and the other shows it as earning above the tolerance band), the conflict should be resolved by treating the class the same as one that falls within the tolerance band.

To help explain and clarify these recommendations, I prepared an illustrative analysis, based upon the Companies' proposed revenue requirements initially filed in these cases. Exhibit___(DP-1) shows the effects of applying boundaries of 0.75 and 1.25 (KEDNY) and 0.5 and 1.5 (KEDLI) times the system average percentage increase to the revenue allocation process in these rate cases. However, I recognize that somewhat different boundaries might be appropriate if the approved revenue increases are significantly different from the

amounts requested by the Companies in their initial testimony. Thus, I recommend the Commission set specific boundaries around the amount of base delivery revenue change applied to individual SCs to reflect the magnitude of the delivery revenue increase that is ultimately adopted by the Commission in these proceedings.

V. RATE DESIGN

A. *Background*

Q. How many firm gas residential accounts are in the Companies' residential Service Classes and what is the difference between residential heating and non-heating customers?

A. Residential customers are the majority of the Companies' accounts and generate 69% (KEDNY) and 67% (KEDLI) of the base delivery revenues; these customers utilize 49% (KEDNY) and 54% (KEDLI) of the system deliveries.[47] There are 1,218,969 residential accounts in the KEDNY service territory; of these accounts 594,542 are non-heating (48.8%) and 624,427 accounts are heating (51.2%). For KEDLI, there are 543,240 KEDLI residential accounts; of these accounts 93,819 are non-heating (17.3%) and 449,421 are heating (82.7%). While the tariffs include a broader description, SC 1B residential heating customers generally use natural gas for space and/or central heating equipment. SC 1A residential non-heating customers are generally customers who only use gas for purposes other

[47] See allocators set forth in Exhibit___(RDP-3), Schedule 8A for both KEDNY and KEDLI.

than space heating and/or central heating such as for gas stovetop cooking, gas dryers, and/or gas water heaters.

Q. What changes are the Companies proposing for the majority of its firm sales residential gas rates?

A. The Companies propose to increase the current residential SC 1A non-heating and SC 1B heating monthly gas customer charges, which includes the first three therms of usage, as follows:[48]

Table 5. Companies' Current and Proposed Residential Customer Charges

Residential Service Class	*Current Customer Charges*[49]	Company *Proposed Customer Charges*
KEDNY		
SC 1A non-heating	$16.25	$19.50
SC 1B heating	$21.55	$24.25
KEDLI		
SC 1A non-heating	$19.75	$22.30
SC 1B heating	$21.66	$23.00

The remaining delivery revenue increase for these classes will be collected through the volumetric portion, which consists of two block rates (Block 2 and Block 3). These two volumetric block rates are designed such that the Block 2 rates are

[48] *See* Exhibit___(RDP-4), Schedule 4 for each Company.

[49] The current customer charges in these 2019 cases were based on the Commission approval of the 2016 Joint Proposal in cases 16-G-0058 *et al*. where residential non-heating customers received an increase in each rate year (KEDNY customer charge increased from $13.74 to $14.70 (RY1), $15.54 (RY2), $16.25 (RY3)) and (KEDLI customer charge increased from $17.65 to $18.53 (RY1), $19.18 (RY2), $19.75 (RY3)). Further, in the 2016 Joint Proposal, customer charges for heating customers were held constant (KEDNY $21.55 and KEDLI $21.66).

higher rate than Block 3 tail-block rates – creating declining block rate structures. As shown in the tables below, there are no significant changes from the current and proposed ratios of tail-block rate (Block 3) to mid-block rate (Block 2). For comparison purposes, a flat rate structure would have a ratio of 100% and an inclining rate structure would have a ratio greater than 100%.

Table 6. Companies' Current and Proposed Tail-Block Rate to Mid-Block Rate Ratio

Block 3/ Block 2 Ratio	Company *Current*	Company *Proposed*
KEDNY		
SC 1A Non-heating	45%	50%
SC 1B Heating	35%	38%
KEDLI		
SC 1A Non-heating	24%	24%
SC 1B Heating	23%	22%

The Companies' choice to maintain the residential declining volumetric block rate structures was based, in part, on its conclusion that it would not be acceptable to switch from a declining block rate to an inclining block rate. Pursuant to the Order Adopting Terms of Joint Proposal and Establishing Gas Rate Plans (issued and effective December 16, 2016) in cases 16-G-0058 *et al*., the Companies were required to conduct a bill impact study. The Companies' inclining block analysis can be found in each of the Companies' Exhibit___(RDP-11) to the Rate Design Panel. The Companies' bill impact study conclusions are as follows:

> The results illustrated by the study – widely divergent percentage increases among customer groups within a class, and huge burdens on lower and/or higher usage customers – are not acceptable from ratemaking perspective. The Panel notes that if the customer charge were maintained at the

> present level, the effect for both SC 1A and SC 1B would be exacerbated. (KEDNY Rate Panel, p. 70)
>
> and
>
> Neither of the results illustrated by the study – regressive increases or a huge burden on higher usage customers – are acceptable from a ratemaking perspective. The Panel notes that if the customer charge were maintained at the present level, the effect for SC 1A would be somewhat mitigated, but the effect for SC 1B, which has almost four times the revenue and twelve times the volume of SC 1A, would be exacerbated. (KEDLI Rate Panel, p. 71)

The Companies ultimately choose to apply modest changes to the volumetric declining rates (*e.g.*, slightly change the ratio of tail-block to mid-block) in combination with increasing the customer charges. See below for the Companies' specific proposed rates and average bill impacts extracted from Exhibit___(RDP-4), Schedule 4.

Table 7. KEDNY's Current and Proposed Delivery Rates and Proposed Average Monthly Bill Increase

KEDNY *Residential Service Class*	*Delivery Rate Structure*	*Current Delivery Rates*	*Proposed Delivery Rates*	*Average Monthly Bill Increase*
SC 1A non-heating	Block 1: Customer Charge	$16.25	$19.50	$7.70 or 25.53% (delivery)
	Block 2: Per Therm Rate (4-50 therms)	$1.7273	$2.1900	$7.72 or 21.62% (total bill)
	Block 3: Per Therm Rate (Over 50 therms)	$0.7690	$1.1031	At 10 therms
SC 1B heating	Block 1: Customer Charge	$21.55	$24.25	$16.50 or 17.78% (delivery)
	Block 2: Per Therm Rate (4-50 therms)	$1.1429	$1.3200	$16.66 or 11.99% (total bill)
	Block 3: Per Therm Rate (Over 50 therms)	$0.4050	$0.5029	At 83 therms

Table 8. KEDLI's Current and Proposed Delivery Rates and Proposed Average Monthly Bill Increase

KEDLI *Residential Service Class*	*Delivery Rate Structure*	*Current Delivery Rates*	*Proposed Delivery Rates*	*Average Monthly Bill Increase*
SC 1A non-heating	Block 1: Customer Charge	$19.75	$22.30	$6.13 or 12.97% (delivery)
	Block 2: Per therm rate (4-50 therms)	$1.4354	$1.5600	$6.28 or 10.76% (total bill)
	Block 3: Per therm rate (Over 50 therms)	$0.3407	$0.3717	At 21 therms
SC 1B Heating	Block 1: Customer Charge	$21.66	$23.00	$6.53 or 6.92% (delivery)
	Block 2: Per Therm rate (4-50 therms)	$1.2839	$1.3500	$7.14 or 5.15% (total bill)
	Block 3: Per Therm rate (Over 50 therms)	$0.2938	$0.3003	At 83 therms

While each of the Companies' Rate Design Panel Exhibit___(RDP-4) shows the bill impacts for customers using 0 to 100 therms (SC 1A Non-heating) and 0 to 400 therms (SC 1B Heating) a month, these exhibits do not display the entire range of bill impacts for various customer usage levels within those service classes. I note that the exhibits do include the proposed typical bill for the average customers in those classes. However, as illustrated in Exhibit___(DP-2) UIU IR No. 11 Attachment 1, there is a wide range of residential customers in each billing usage range (this IR shows number of customers at each usage level in KEDNY and KEDLI residential SCs in July and January. Additionally, Exhibit___(DP-2) PULP IR No. 95 Attachment 1 shows the residential data broken down by heating and non-heating for both Companies. The maximum monthly usage shown in these IRs range from 25,000 to over 50,000 therms (including low income customers).

B. ***Recommendations***

Q. What is your response to the Companies' proposed increases in the customer charges and volumetric block rate changes for residential accounts?

A. The starting point for my recommended rate design is my proposed revenue allocation, which I used to assign costs to individual service classes. There are a few variables that impact rates such as: (1) revenue requirement, (2) revenue allocation, (3) customer charges, and (4) the portion of the revenues that flow through each of the volumetric rates (*e.g.*, increasing the ratio of tail-block to mid-block). This discussion is explanatory in nature, since UIU's recommendation will be dependent upon the final revenue requirement for each utility, which will influence the magnitude of any potential changes to the customer charge and volumetric rates.

Residential Non-Heating – SC 1A

The Companies' Residential Non-Heating SC 1A classes are very sensitive to any rate adjustments. I recommend moving away from the existing, steeply declining rate design. However, the movement must be gradual. A moderate change is needed to not financially overburden high usage customers (including some low-income customers). However, if the Companies' residential SC 1A class revenue responsibilities are reduced as per my revenue allocation proposals and/or if the overall revenue requirements are reduced in general, then there should be greater movement toward flat rates (especially for KEDLI since KEDLI SC 1A has a lower

tail-block to mid-block ratio than KEDNY as well as a lower tail-block rate). For the higher users (*e.g.*, customers who use greater than 500 therms per month while typical customers use 10-21 therms), the Companies should also check to make sure these customers are in fact non-heating customers, and move them to the correct class, where applicable.

Residential Heating – SC 1B

The Companies' residential SC 1B heating rates should be a designed to more effectively encourage energy efficiency improvements. As such, the SC 1B heating rates should, at a minimum, be gradually moved away from a declining volumetric rate toward a flat rate. This recommendation contrasts with the Companies' proposal to continue a declining rate structure for SC 1B with a similar tail-block rate (Block 3) to mid-block rate (Block 2) ratio, because that approach makes no movement toward encouraging greater energy efficiency. The Companies' SC 1B tail-block rates are lower than the Companies' SC 1A tail-block rates, but in both cases careful consideration must be made on how far the tail-block rates can be increased before unacceptably severe bill impacts affect the high users in these classes. As such, the magnitude of my proposed change will be dependent upon the final revenue requirements.

Residential Bill Impacts

I recommend the residential bill impact results for KEDNY and KEDLI include additional information to account for customers at a wider range of usage levels.

Residential bill impact results for both KEDNY and KEDLI (*e.g.*, Exhibit___(RDP-4)) should be evaluated for the entire range of usage levels of those classes (*e.g.*, usage up to 50,000 therms) and include the percentage of customers (including low income customers) within each usage level so parties can understand the range of bill impacts for those entire SCs. This is particularly helpful analysis for the parties consider how the rate design deviates from the Companies' rate proposals.

Q. Do you have any other comments related to the Companies' proposed rate design?

A. Yes. I recommend the Companies' continue[50] to provide an increased level of targeted assistance to residential customers with the highest monthly gas usage (including low income customers) to ensure they understand how they can reduce their usage. To achieve the State's goals for energy efficiency, these customers need to understand the options they have for weatherization of their home or business, replacing an older furnace with a newer, more efficient one, installing programmable thermostats, and other actions they can take to reduce their monthly utility bill.

[50] *See* Exhibit___(DP-2) (Company response to UIU IR No. 44) (stating "(t)he Companies target high usage gas residential customers through the Gas Residential Engagement program where Home Energy Reports (HERS) are sent to customers and identify their high usage (compared to similar households or their own normal usage.").

Q. Does this conclude your direct testimony, which was prefiled with the Commission on August 30, 2019?

A. Yes.